

About

Hi. I'm a design engineer with ten years' experience and a Masters in Electronic and Electrical Engineering, specialising in rocketry and instrumentation, having apprenticed at NPL. I've built and worked within multi-disciplinary teams to design a liquid rocket engine, develop a liquid oxygen pump, instrument the Orbex Prime launch vehicle, and launch – then lose – the highest altitude amateur rocket in the UK. Please see www.tobythomson.co.uk/projects.

Skills

CAD (SolidWorks, OnShape), PCB design (Altium, KiCad, PADS), bare-metal C (STM32), Python, Linux, GD&T, manual machining, SMT soldering, cryogenic instrumentation, RF fundamentals, technical leadership

Experience

Project Lead, Gryphon II Block II, Leeds University Rocketry Association 2025 – 2026

- Led a four-person team to build and launch an altitude record-breaking rocket: 48 kft apogee, Mach 2.3, largest rocket motor procured by a UK amateur.
- Designed parachute recovery system: full redundancy, bespoke pyrotechnic rope cutters, inertial and GNSS telemetry, RDF tracking.
- Co-ordinated airspace with the CAA and MOD: built the Leeds Flight Simulator for trajectory modelling, authored detailed flight safety assessment.
- Proposed and delivered on a £13k budget.

Avionics Intern, Orbex 2024 – 2025

- Developed solution for in-flight cryogenic propellant gauging: found gaps in the initial validation approach, won approval for a differential-pressure alternative.
- Designed and qualified an engineering camera: vibration and thermal vacuum qualification as per NASA GEVS.
- Liaised across departments to specify and procure an economical data acquisition system and instrumentation for environmental monitoring, saving £50k.
- Earned a four-month extension and a seat on the S1 Integrated Project Team, owning Prime's first-stage design decisions.

Chief Engineer, Leeds University Rocketry Association 2022 – 2024

- Initiated and led the development of the Gryphon II sounding rocket: architected the vehicle, owned £20k budget and timeline, designed the avionics bay and "universal coupler" joint.
- Mechanical design of a LOX/IPA rocket engine, 2 kN design thrust, tested at Race to Space 2023.
- Produced a training wiki for the 30+ members and reviewed all technical work, from PCBs to firmware and CAD.
- Fundraised a recurring £25k for LURA's activities, via the university and direct corporate sponsorship.

Avionics Engineer, Leeds University Rocketry Association 2021 – 2022

- Built avionics and recovery systems for Gryphon I and I-C: avionics barometric testing, developed piston-actuated recovery system to reduce black-powder use by 100x: runner-up, 2022 Spaceport America Cup Innovation Award, 4th in category.
- Introduced a house drawing standard conforming to BS8888, implemented technical document control.

Research Engineer, National Physical Laboratory 2020 – 2021

- Developed the STAR-CC-OGSE Monitor Detector, from concept to completion: used in calibrating the MicroCarb satellite. Vacuum-resident, thermally managed, custom optomechanics and analogue read-out. Wrote the interface spec, reverse-engineered an undocumented spectrometer, led mechanical and electrical design, project managed.

Engineering Apprentice, National Physical Laboratory 2016 – 2020

- Designed and built bespoke equipment for long-term experiments and commercial measurement services, including sensitive Kibble Balance work.
- Rotations in mechanical design, PCB design, optics, electrical installation and manual machining. Some cryogenic handling.

Education

<i>MEng Electronic and Electrical Engineering (First Class), The University of Leeds</i>	2021 – 2026
• Master's project: <u>liquid oxygen pump</u>, hand-built brushless motor, 3 kg/s at 40 bar. • Bachelor's project: pure-C <u>GPS spoofing tool</u>.	
<i>HNC Electronic and Electrical Engineering (Distinction), Kingston College</i>	2018 – 2020
• Final project: <u>RF transceiver</u> for PocketQube satellite, 1.5 W TX at 434 MHz.	
<i>BTEC Level 3 Extended Diploma in Engineering (D*D*D*), Kingston College</i>	2016 – 2018

Awards

<i>Race to Space, Innovation Award</i> , for early amateur work in electric turbomachinery with <u>Zola</u> .	2025
<i>Race to Space, Best Team Spirit</i> , for sustained morale through a demanding test campaign.	2023
<i>Whitworth Scholarship</i> , for demonstrated academic and practical engineering ability.	2021